

North Star

Contents

1 2026.07.06 - North Star: the virtual cell, and how torchcell differs	1
1.1 The consensus (Qian et al. 2026, Nature Perspective “Towards the construction of a virtual yeast”) . . .	1
1.2 Shared goals (state briefly, don’t over-tie)	1
1.3 Where torchcell is DIFFERENT (for the manuscript)	1
1.4 Honest complementarity (keep the positioning credible)	2
1.5 For the paper – lead with these three	2

1 2026.07.06 - North Star: the virtual cell, and how torchcell differs

A shared reference for consistent positioning in the manuscript (may not compile into the final paper). Anchors the field’s consensus goal and, more importantly, **how torchcell is different** – since the consensus paper itself claims to be biology-first, our differentiation must be sharper than that. Background memory: **virtual-cell-consensus-goal**.

1.1 The consensus (Qian et al. 2026, Nature Perspective “Towards the construction of a virtual yeast”)

Authored by much of the yeast-systems establishment (Costanzo, Boone, Andrews, Ralser, Steinmetz, Zampieri, Cai, Yue, Schacherer, Weinan E, Gao, Guo, ...). Proposes an **AI-driven virtual cell (AIVC)** for *S. cerevisiae*: the cell decomposed into **8 function-centred modules** (membrane, genetic hubs, mitochondrial, cytosolic metabolism, biosynthetic, cytoskeletal, stress, degradation), each a domain AI tool, coordinated by an **LLM orchestration layer**, grounded by **knowledge graphs + ontologies + mechanistic verifiers (RL from verifiable rewards)**. Three **data pillars**: a priori knowledge, static spatial architecture (cryo-ET, expansion/spatial omics, GFP localization), dynamic multi-omics after perturbation. A closed-loop active-learning flywheel (**WAY-AL, argmax expected information gain**) with robotic multi-omics generates new data (969 strains -> a core set; >15,000 time-resolved proteomes; >5,000 metabolomes). Prototype = the metabolic / synthetic-biology module (yeast-GEM $\$ \cdot \$ v=0$ + paired proteome/metabolome). Explicitly a 5-10 yr consortium (biology / data / AI / applications cores); NOT a digital twin.

Crucially, they already frame themselves as biology-first: “our perspective emphasizes starting from cellular function, experimental data and testable hypotheses, with AI serving as an integrating engine.” So “we are biology-first” is NOT a differentiator.

1.2 Shared goals (state briefly, don’t over-tie)

Virtual cell as the north star; yeast first; mechanistic (GEM) grounding; perturbation -> phenotype as the core relation; ontologies/knowledge graphs; a synthetic-biology metabolic prototype. Their dynamic-states pillar draws on largely the same experimental corpus torchcell schematizes (genetic interactions, morphology, expression, proteome, amino-acid metabolome). Common ground is broad – avoid leaning on any single dataset coincidence.

1.3 Where torchcell is DIFFERENT (for the manuscript)

The difference is **layer, rigor, and operationalization**, not philosophy.

- 1. Substrate vs. modeling stack (the layer).** They build the predictive / generative / active-data-generation stack (functional AI modules + generative models + robotic experimentation). torchcell builds the **experimental-data ONTOLOGY + substrate** those models must stand on. Complementary layers – but the substrate is under-addressed, and a virtual cell is only as trustworthy as the data ontology beneath it.
- 2. Provenance-first, rebuildable, VERIFIED (their blind spot – the strongest point).** The consensus paper is silent on data provenance, versioning, rebuild guarantees, and record-level validation. It consumes “curated databases” as given. torchcell makes every value trace to a **sha256**-pinned artifact + verbatim source quote, with a rebuild guarantee and **L0-L4 verification** (structure / count / value-fidelity / reference / cross-source). Their own listed challenges – “integrating experimental measurements with knowledge bases introduces uncertainty,” genetic-background effects (they cite 18.5% of KO phenotypes varying by background), incomplete variant-level coverage – are symptoms of exactly the untyped, unversioned, un-provenanced substrate we replace.
- 3. Ontologize the EXPERIMENT, not a presupposed decomposition of the cell.** They impose a top-down 8-module functional taxonomy of the cell. We model the **experiment**: a typed, validated (**genotype $\times$ environment**) -> **phenotype** record with perturbations, provenance, and phenotype-composition semantics (one phenotype per dataset $\times$ measurement modality). Structure EMERGES from accumulated verifiable records and can be projected into any downstream taxonomy (including theirs). This is robust to new data types and avoids baking in a possibly-wrong functional decomposition. Our ontology is the **AUTHORITATIVE** representation; theirs is a soft prior nudging a generative model.
- 4. Sequence-level genotype fidelity -> strain design.** They represent genotype as genome embeddings (Evo2)

+ perturbation labels. We model each perturbation as an EDIT to the **total genomic content in the cell** (deletions, integrations, alleles, present episomal plasmids, heterologous cassettes) – the true modified sequence, captured at SBOL/GenBank rigor (`[[torchcell.sequence.plasmid-and-genomic-content-design]]`, `[[torchcell.datamodels.gene-addition-perturbation-design]]`). Their own synthetic-biology / inverse-strain-design goal **REQUIRES** this and leaves it abstract: design means manipulating actual sequence, and predicting on an engineered strain needs the engineered sequence, not WT + a token.

5. **Faithful capture of the primary record, more rigorous than the source.** We go to the paper/SI and record what was actually done to the bit/byte (n, uncertainty TYPE, units, background, environment; back-solving when a value isn't a released column) rather than trusting an aggregated database entry. This is precisely what their acknowledged background/coverage problems demand.
6. **Application + generalization axis.** Their axis is yeast -> other eukaryotes / human (disease, drug discovery, fundamental cell biology). Ours is yeast -> other single-cell **bioproduction hosts** – yeast as a chassis for inverse strain design (heterologous pathways, plasmids, CBM metabolite yield). Yeast is first and the manuscript target.

1.4 Honest complementarity (keep the positioning credible)

We do NOT claim to replace their modeling / generative / spatial / active-learning stack; several things they emphasize (3D spatial architecture, generative simulation, autonomous data generation) are outside our current scope. Our contribution is the **trustworthy, verifiable, sequence-faithful data foundation** their program silently assumes exists. Frame as a complementary foundation, not a competitor.

1.5 For the paper – lead with these three

1. Provenance-first, rebuildable, L0-L4-verified experimental-data ontology (their gap).
2. Ontologize the experiment (typed genotype $\times$ environment $\rightarrow$ phenotype), not a top-down cell decomposition.
3. Sequence-level genotype fidelity (perturbation as edit to total genomic content) as the enabler of inverse strain design.